

LAB 7 – DIODES

Introduction

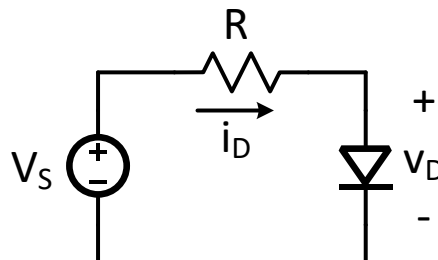
In this lab, you will explore some diode characteristics. Most of the measurements will be done with our usual lab equipment, however, everyone will get a chance to use the semiconductor parameter analyzer to measure the current-voltage characteristics of a couple of diodes.

Since there is only one analyzer, we will have to take turns using it. Each group can have up to 30 minutes to complete measurements on their diodes, which should be more than enough time.

Experiment 1. Diode I-V Characteristics with the Parameter Analyzer

1. Measure the forward characteristics for the following diodes: 1N4006, 1N4148, Red LED, and Green LED. Use the tutorial as a guide and keep in mind that most of the silicon diodes will have similar behaviors. The current should reach the 100 mA limit for forward voltages of approximately 1.25 V or less. The LEDs, however, will require larger forward voltages, in the range of 2 to 3 V, to reach the 100 mA limit.
2. Create and make a copy of a single plot that has the forward I-V characteristic (both linear and log scales) of the 1N4006 diode and the red LED. You will need to use the append measurement function on the parameter analyzer – see the tutorial. To view enough of the plot for a good comparison, the applied voltage should range from 0 to +5 V and the current axis should extend to 100 mA. Include this plot in your lab report.
3. Measure the reverse characteristics (linear scale only) for all of the diodes. The reverse voltage should extend from 0 to -15 V.
4. For the report, create and make a copy of a single plot that has the reverse I-V characteristics of the 1N4006 diode. The measurement should range from 0 to -15 V and the current axis should range from 0 to 20 mA. Make the plot using *linear* axis and include this in your lab report.

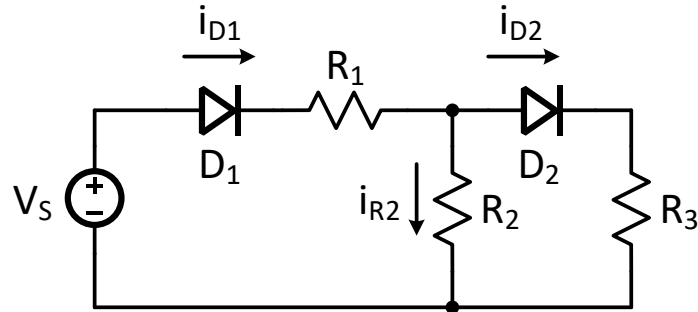
Experiment 2. Diode I-V Characteristics at the Lab Bench



1. Build the circuit shown above using the 1N4006 diode and a $470\ \Omega$ resistor from your lab kit. Carefully measure the resistor value beforehand.
2. Apply the various DC voltages for the source listed in the table below. For each source voltage, measure the corresponding diode voltage and calculate the diode current.
3. Include this data in your lab report.

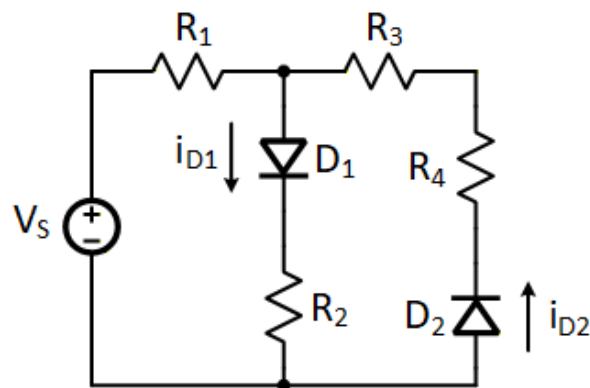
V_S (V)	v_D (V)	i_D (mA)	V_S (V)	v_D (V)	i_D (mA)
-10			+3		
-8			+4		
-6			+5		
-4			+6		
-2			+7		
+0			+8		
+1			+9		
+2			+10		

Experiment 3. Simple Diode Circuits



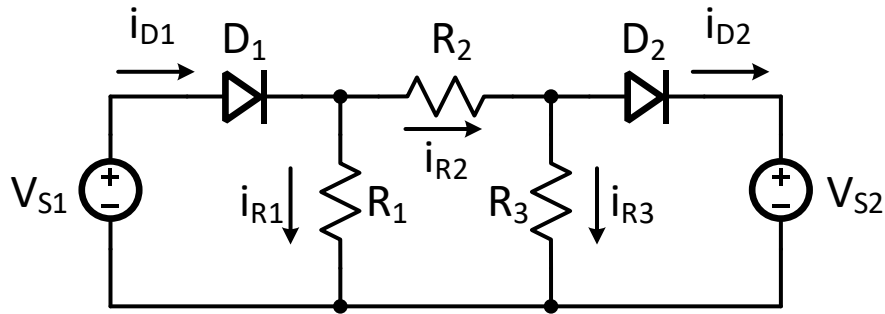
Component	Value
R_1	$1\text{ k}\Omega$
R_2	$2.2\text{ k}\Omega$
R_3	$3.3\text{ k}\Omega$
D_1, D_2	1N4006

1. Calculate the currents, i_{D1} , i_{D2} , and i_{R2} using the constant voltage drop model for the diodes with a 0.7 V drop. Perform the calculations assuming $V_S = 1.0\text{ V}$ and 5.0 V .
2. Build the circuit shown in the schematic above. All component values are shown in the table.
3. Set the source voltage to 1.0 V and measure each current, i_{D1} , i_{D2} , and i_{R2} .
4. Repeat the measurements with $V_S = 5.0\text{ V}$.



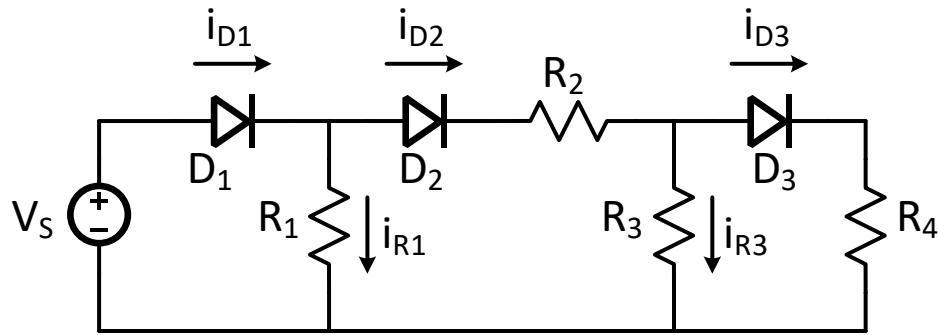
Component	Value
R_1	$1\text{ k}\Omega$
R_2	$1\text{ k}\Omega$
R_3	$2.2\text{ k}\Omega$
R_4	$4.4\text{ k}\Omega$
D_1, D_2	1N4006

5. Calculate the currents, i_{D1} and i_{D2} using the constant voltage drop model for the diodes with a 0.7 V drop. Perform the calculations assuming $V_S = 5.0$ V and -5.0 V.
6. Build the circuit shown in the schematic. All component values are shown in the table.
7. With $V_S = 5.0$ V, measure the currents i_{D1} and i_{D2} .
8. Repeat the measurements with $V_S = -5.0$ V.



Component	Value
R_1	1.5 k Ω
R_2	2.2 k Ω
R_3	4.7 k Ω
D_1, D_2	1N4006

9. Calculate the currents, i_{D1} , i_{D2} , i_{R1} , i_{R2} , and i_{R3} using the constant voltage drop model for the diodes with a 0.7 V drop. Assume that $V_{S1} = 10$ V and $V_{S2} = 5$ V.
10. Build the circuit shown in the schematic above and set $V_{S1} = 10$ V and $V_{S2} = 5$ V. All other component values are shown in the table.
11. Measure the currents, i_{D1} , i_{D2} , i_{R1} , i_{R2} , and i_{R3} .



Component	Value
R_1	$1.5\text{ k}\Omega$
R_2	$2.2\text{ k}\Omega$
R_3	$1.5\text{ k}\Omega$
R_4	$1\text{ k}\Omega$
D_1, D_2, D_3	1N4006

12. Calculate the currents, i_{D1} , i_{D2} , i_{D3} , i_{R1} and i_{R3} using the constant voltage drop model for the diodes with a 0.7 V drop. Perform the calculations assuming $V_S = 1.0\text{ V}$ and 5.0 V .
13. Build the circuit shown in the schematic above. All component values are shown in the table.
14. Set $V_S = 1.0\text{ V}$ and measure all currents.
15. Repeat the measurements with $V_S = 5.0\text{ V}$